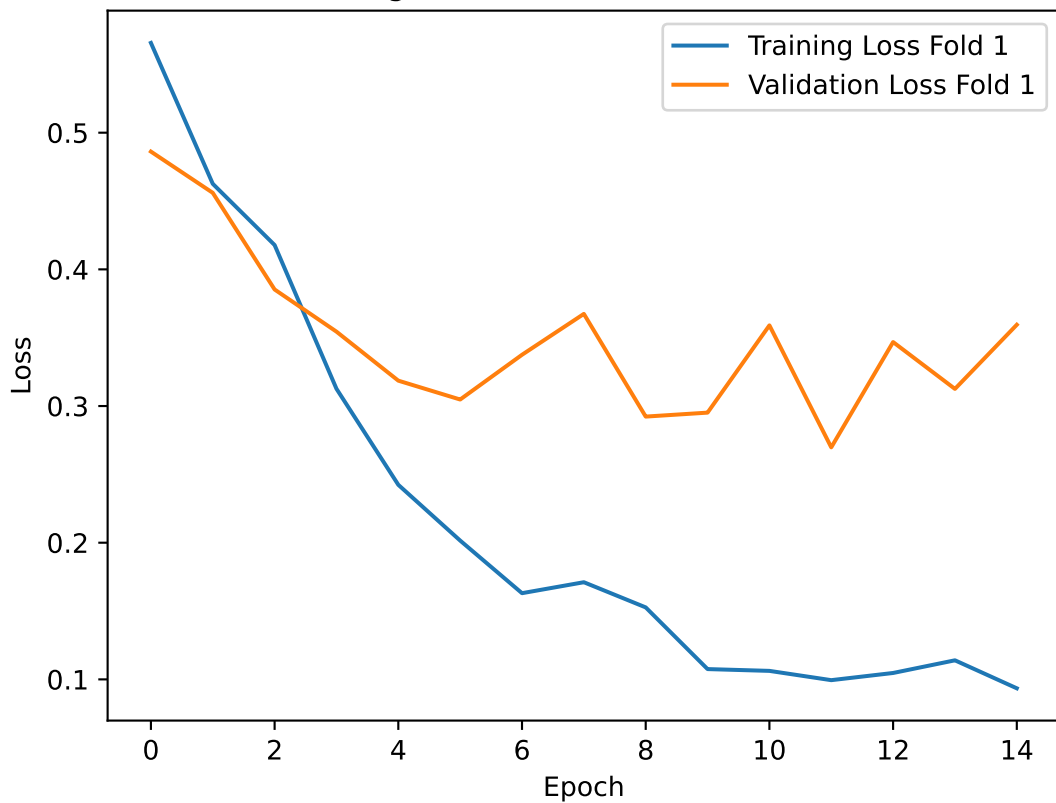
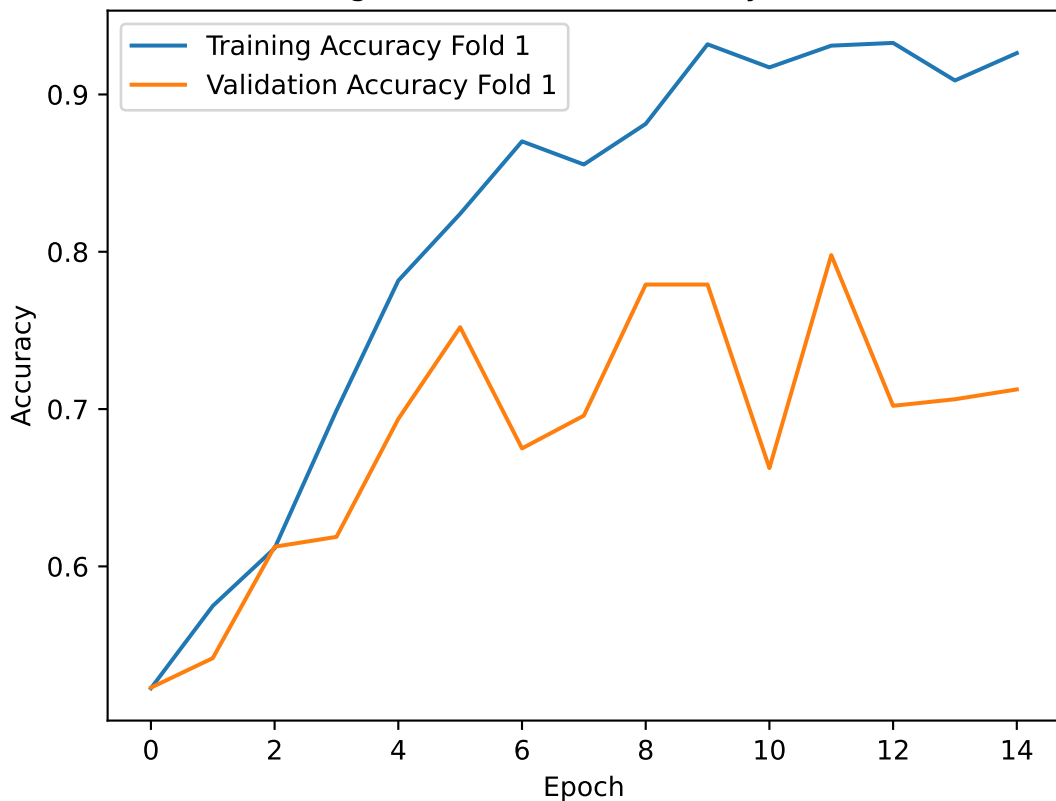


Training and Validation Loss for Fold 1



Training and Validation Accuracy for Fold 1



Best Hyperparameters for Fold 1:

```
{'conv_filters': 96, 'conv_activation': 'relu', 'unitsD1': 160, 'unitsD2': 32}
```

Best Trial Score for Fold 1: 0.8145833015441895

https://storage.googleapis.com/tensorflow/keras-applications/resnet/resnet50_weights_tf_dim_ordering_tf_kernels_notop.h5

```
0s/step [1m 49152/94765736 [0m
```

```
4us/step[0m 81920/94765736[0m
```

5us/step [1m 131072/94765736] [0m
[37m [0m [1m5:23[0m

3us/step [0m 163840/94/65/36] [0m
[37m [0m [1m4:55 [0m

30s/step [0m 1m 212992/94765736 0m
[37m [0m [1m4:16 0m

33s/step [1m 202144/94765750] [0m]
[37m [0m [1m3:49[0m

[illegible]

2us/step [0m 425984/94/65/36] [0m
[37m [0m [1m 2:45 [0m

```
2us/step[1m 524288/94765736[0m
```

```

[57m-----[0m [1m2.24[0m
2us/step[0m [1m 638976/94765736[0m

```

```

[37m-----[0m [1m2:06[0m
1us/stop[1m 786432/94765736[0m

```

037m 0m 1m1:48 0m

[illegible]

```
1us/step [0m 1228800/94765736] [0m
```

```
1us/step
```

```
[37m=====[0m [1m1:05[0m  
1us/step#####[1m 1867776/94765736[0m
```

```

[37m-----[0m [1m55s[0m 1us/step
#####1m 2324720/04765736[0m

```

[37m [0m [1m46s [0m

0us/step [0m 2924544/94/65/36] [0m
[37m [0m [1m 38s [0m

```
0us/step [1m 3645440/94765736] 0m
```

```
0us/step
```

```

0us/step

```

00:32m-00:37m-----00:00 01m21s00:00

ods/step [1m 09/1392/94/65/30] [0m
[32m-[0m [37m [0m [1m17s [0m

```
0us/step [1m 8413184/94765736] [0m
[32m-[0m [32m-[0m [1m15s [0m
```

```
0us/step
```

```
0us/step
```

```
00:32m--00:37m-----00m 01m12s00m
00:32m--00:37m-----00m 01m12s00m
```

[32m—[0m [37m—————[0m [1m11s [0m

0us/step [1m13s156352/94765736] [0m
[32m=[0m [37m=[0m [1m10s [0m

```

[32m[0m[37m[0m[1m9s[0m 0us/step

```

[illegible]

0us/step

□[32m□[0m□[37m□[0m□[1m8s□[0m

ods/step [1m19128520/94765756] [0m
[32m [0m [37m [0m [1m7s [0m

0us/step [1m20242432/94765736] [0m
[32m [0m [32m [0m [1m3c [0m

```
0us/step
```

0us/step

```

[32m[0m[37m[0m[1m6s[0m
0us/step[1m33863296/94765736[0m

```

□[32m—□[0m□[37m—□[0m □[1m6s□[0m

0us/step [1m23100288/9476536] [0m
[32m [0m [37m [0m [1m5s [0m

```
0us/step[0m
[32m[0m[32m[0m[1m5s[0m
```

```
0us/step
```

0us/step

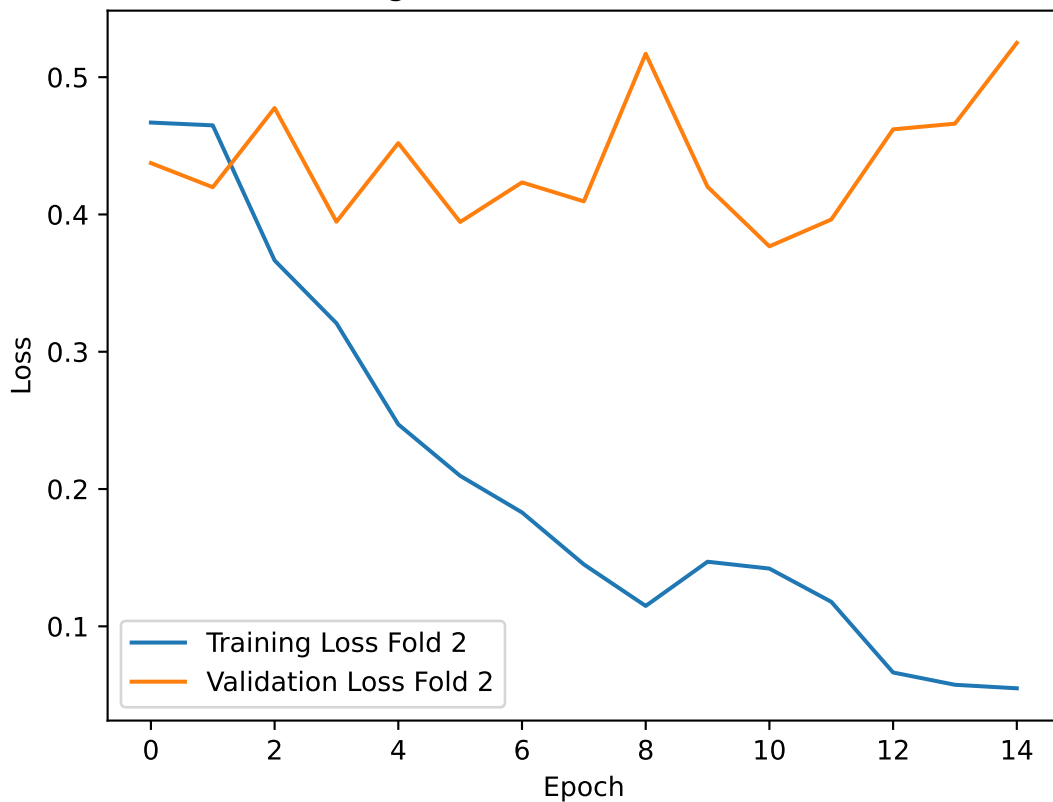
0x[32m[0m[37m[0m[1m5s[0m

0m 32m 0m 37m 0m 1m5s 0m

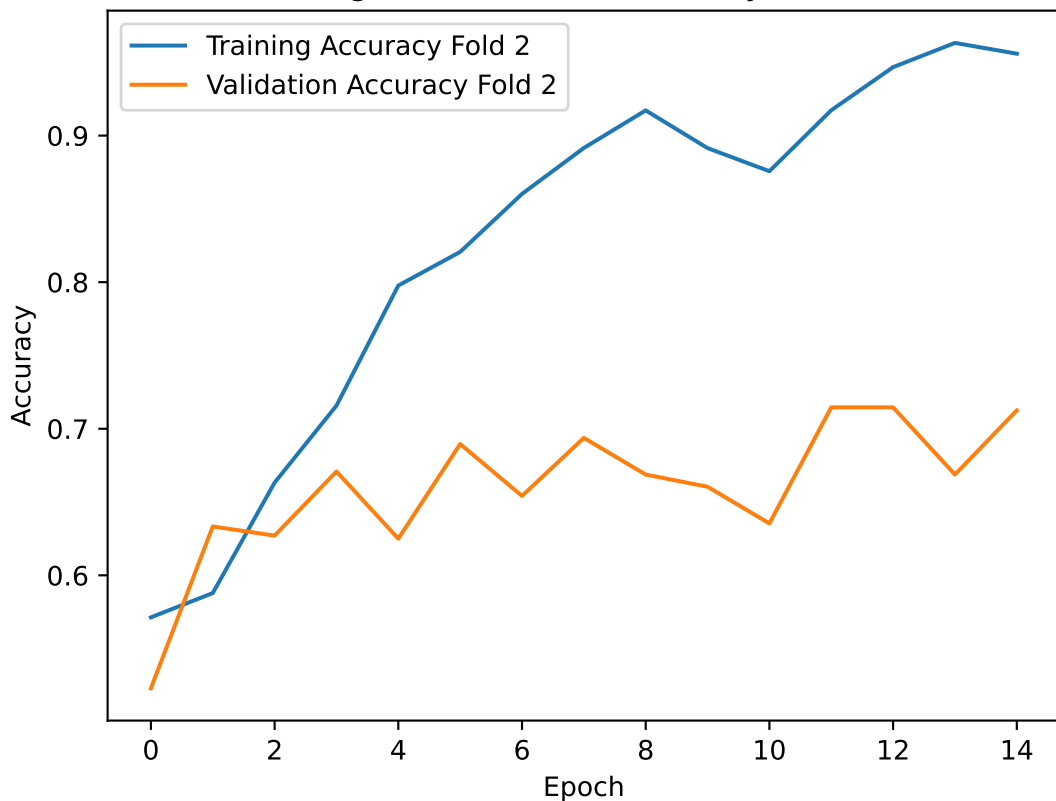
0us/step [1m3145/280/94/65/36] 0m
[32m [0m [37m [0m [1m4s [0m

```
0us/step[1m32735232/94765736[0m
```

Training and Validation Loss for Fold 2



Training and Validation Accuracy for Fold 2



Best Hyperparameters for Fold 2:

`{'conv_filters': 64, 'conv_activation': 'relu', 'unitsD1': 288, 'unitsD2': 32}`

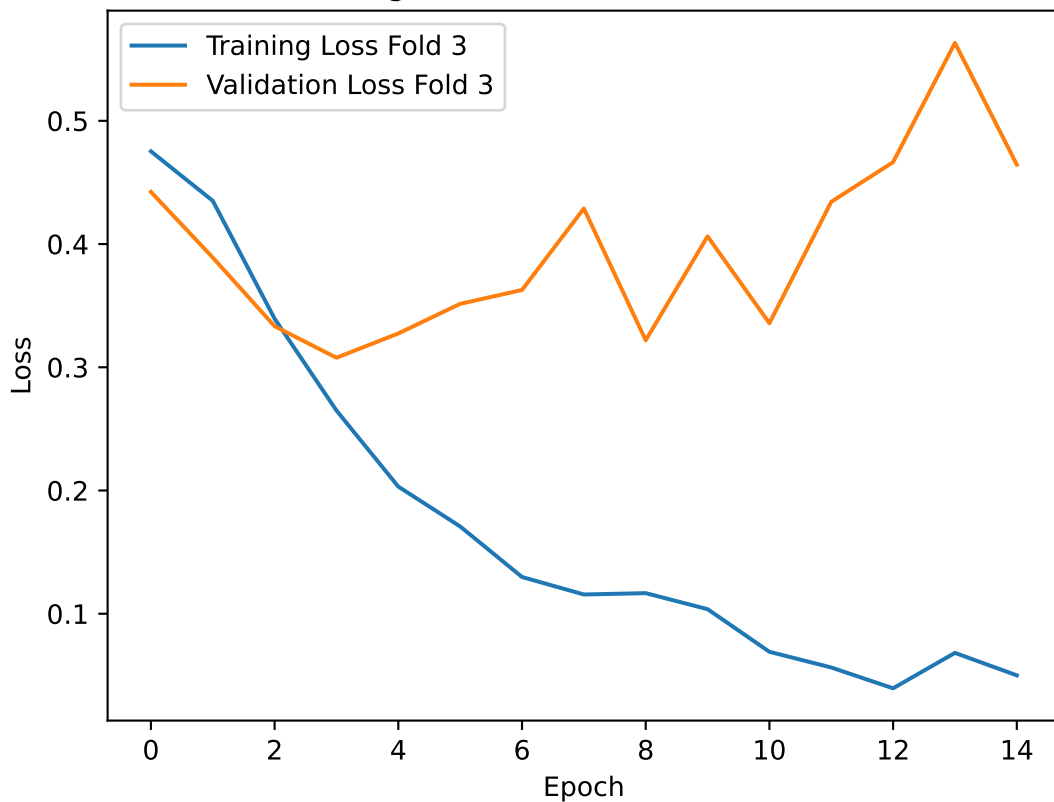
Best Trial Score for Fold 2: 0.7583333253860474

Search: Running Trial #1

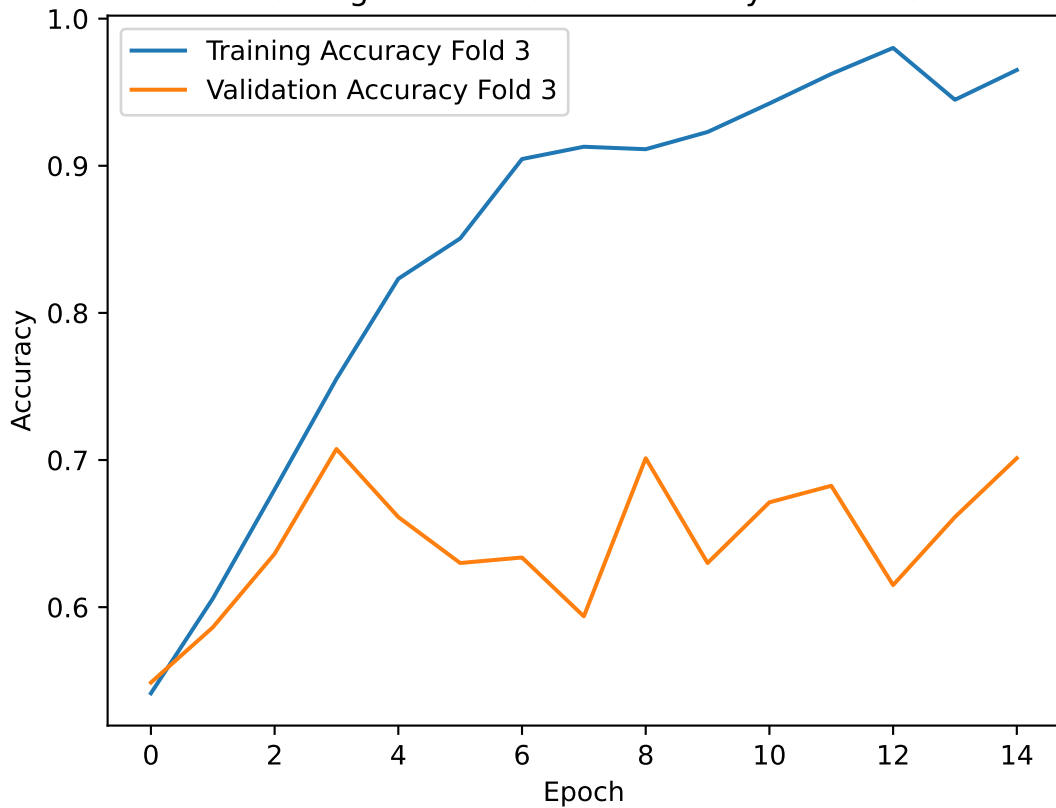
Value	Best Value So Far	Hyperparameter
256	256	conv_filters
relu	relu	conv_activation
32	32	unitsD1
32	32	unitsD2

Epoch 1/15
[1m 1/17][0m [32m][0m[37m-----[0m [1m9:59][0m 37s/step - accuracy: 0.3750 - loss: 0.4907
[32m][0m[37m-----[0m [1m3s][0m 241ms/step - accuracy: 0.5000 - loss: 0.4209
[32m][0m[37m-----[0m [1m3s][0m 238ms/step - accuracy: 0.5451 - loss: 0.3973
[32m][0m[37m-----[0m [1m3s][0m 238ms/step - accuracy: 0.5612 - loss: 0.3971
[32m][0m[37m-----[0m [1m2s][0m 239ms/step - accuracy: 0.5640 - loss: 0.4018
[32m][0m[37m-----[0m [1m2s][0m 239ms/step - accuracy: 0.5637 - loss: 0.4086
[32m][0m[37m-----[0m [1m2s][0m 239ms/step - accuracy: 0.5629 - loss: 0.4142
[32m][0m[37m-----[0m [1m2s][0m 239ms/step - accuracy: 0.5624 - loss: 0.4190
[32m][0m[37m-----[0m [1m1s][0m 239ms/step - accuracy: 0.5608 - loss: 0.4232
[32m][0m[37m-----[0m [1m1s][0m 239ms/step - accuracy: 0.5598 - loss: 0.4277
[32m][0m[37m-----[0m [1m1s][0m 239ms/step - accuracy: 0.5590 - loss: 0.4314
[32m][0m[37m-----[0m [1m1s][0m 239ms/step - accuracy: 0.5584 - loss: 0.4342
[32m][0m[37m-----[0m [1m0s][0m 239ms/step - accuracy: 0.5580 - loss: 0.4366
[32m][0m[37m-----[0m [1m0s][0m 239ms/step - accuracy: 0.5573 - loss: 0.4389
[32m][0m[37m-----[0m [1m0s][0m 239ms/step - accuracy: 0.5577 - loss: 0.4407
[32m][0m[37m-----[0m [1m0s][0m 239ms/step - accuracy: 0.5580 - loss: 0.4420
[32m][0m[37m[0m [1m0s][0m 861ms/step - accuracy: 0.5578 - loss: 0.4434
[32m][0m[37m[0m [1m71s][0m 2s/step - accuracy: 0.5576 - loss: 0.4447 - val_accuracy: 0.5667 - val_loss: 0.4398
Epoch 2/15
[1m 1/17][0m [32m][0m[37m-----[0m [1m4s][0m 307ms/step - accuracy: 0.4375 - loss: 0.5350
[32m][0m[37m-----[0m [1m3s][0m 239ms/step - accuracy: 0.5703 - loss: 0.4406
[32m][0m[37m-----[0m [1m3s][0m 238ms/step - accuracy: 0.6024 - loss: 0.4181
[32m][0m[37m-----[0m [1m3s][0m 238ms/step - accuracy: 0.6042 - loss: 0.4328
[32m][0m[37m-----[0m [1m2s][0m 238ms/step - accuracy: 0.5921 - loss: 0.4467
[32m][0m[37m-----[0m [1m2s][0m 239ms/step - accuracy: 0.5854 - loss: 0.4565
[32m][0m[37m-----[0m [1m2s][0m 239ms/step - accuracy: 0.5815 - loss: 0.4631
[32m][0m[37m-----[0m [1m2s][0m 239ms/step - accuracy: 0.5752 - loss: 0.4698
[32m][0m[37m-----[0m [1m1s][0m 238ms/step - accuracy: 0.5688 - loss: 0.4756
[32m][0m[37m-----[0m [1m1s][0m 238ms/step - accuracy: 0.5629 - loss: 0.4805
[32m][0m[37m-----[0m [1m1s][0m 239ms/step - accuracy: 0.5602 - loss: 0.4829
[32m][0m[37m-----[0m [1m1s][0m 239ms/step - accuracy: 0.5585 - loss: 0.4841
[32m][0m[37m-----[0m [1m0s][0m 239ms/step - accuracy: 0.5575 - loss: 0.4849
[32m][0m[37m-----[0m [1m0s][0m 239ms/step - accuracy: 0.5572 - loss: 0.4852
[32m][0m[37m-----[0m [1m0s][0m 239ms/step - accuracy: 0.5577 - loss: 0.4850
[32m][0m[37m-----[0m [1m0s][0m 239ms/step - accuracy: 0.5579 - loss: 0.4847
[32m][0m[37m[0m [1m0s][0m 234ms/step - accuracy: 0.5576 - loss: 0.4845
[32m][0m[37m[0m [1m6s][0m 332ms/step - accuracy: 0.5573 - loss: 0.4843 - val_accuracy: 0.5875 - val_loss: 0.4072
Epoch 3/15

Training and Validation Loss for Fold 3



Training and Validation Accuracy for Fold 3



Best Hyperparameters for Fold 3:

`{'conv_filters': 128, 'conv_activation': 'relu', 'unitsD1': 288, 'unitsD2': 160`

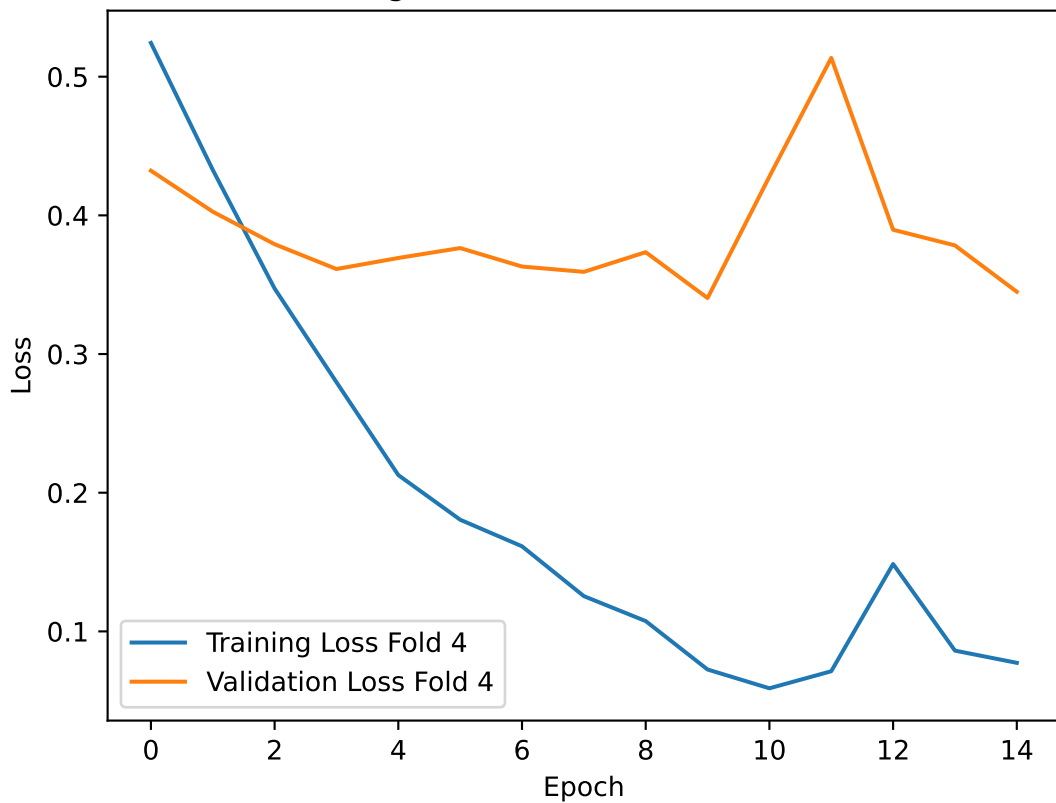
Best Trial Score for Fold 3: 0.7875000238418579

Search: Running Trial #1

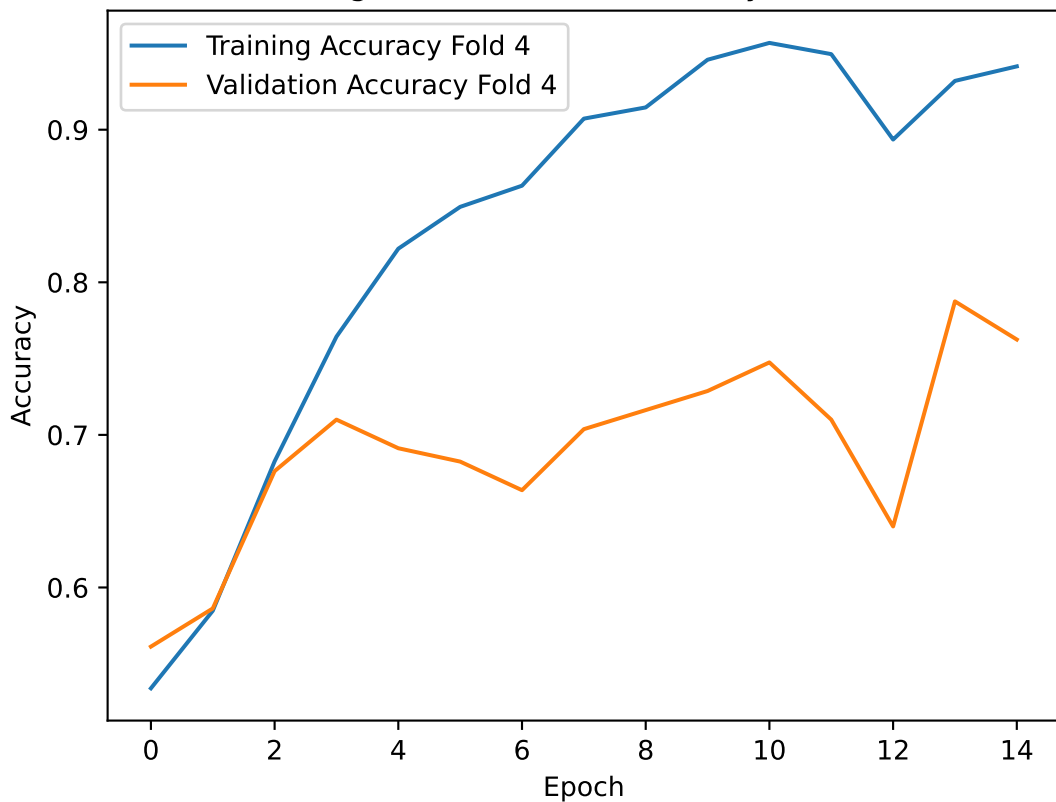
Value	Best Value So Far	Hyperparameter
64	64	conv_filters
relu	relu	conv_activation
160	160	unitsD1
160	160	unitsD2

Epoch 1/15
[1m 1/17][0m [32m-[0m[37m-----[0m [1m10:20][0m 39s/step - accuracy: 0.3750 - loss: 0.5196
[32m-[0m[37m-----[0m [1m3s][0m 247ms/step - accuracy: 0.4922 - loss: 0.4484
[32m-[0m[37m-----[0m [1m3s][0m 245ms/step - accuracy: 0.5538 - loss: 0.4064
[32m-[0m[37m-----[0m [1m3s][0m 245ms/step - accuracy: 0.5658 - loss: 0.4035
[32m-[0m[37m-----[0m [1m2s][0m 245ms/step - accuracy: 0.5714 - loss: 0.4054
[32m-[0m[37m-----[0m [1m2s][0m 245ms/step - accuracy: 0.5690 - loss: 0.4101
[32m-[0m[37m-----[0m [1m2s][0m 245ms/step - accuracy: 0.5681 - loss: 0.4140
[32m-[0m[37m-----[0m [1m2s][0m 246ms/step - accuracy: 0.5659 - loss: 0.4183
[32m-[0m[37m-----[0m [1m1s][0m 246ms/step - accuracy: 0.5625 - loss: 0.4229
[32m-[0m[37m-----[0m [1m1s][0m 246ms/step - accuracy: 0.5571 - loss: 0.4283
[32m-[0m[37m-----[0m [1m1s][0m 246ms/step - accuracy: 0.5532 - loss: 0.4333
[32m-[0m[37m-----[0m [1m1s][0m 246ms/step - accuracy: 0.5505 - loss: 0.4372
[32m-[0m[37m-----[0m [1m0s][0m 246ms/step - accuracy: 0.5479 - loss: 0.4404
[32m-[0m[37m-----[0m [1m0s][0m 246ms/step - accuracy: 0.5464 - loss: 0.4428
[32m-[0m[37m-----[0m [1m0s][0m 246ms/step - accuracy: 0.5450 - loss: 0.4448
[32m-[0m[37m-----[0m [1m0s][0m 246ms/step - accuracy: 0.5446 - loss: 0.4461
[32m-[0m[37m[0m [1m0s][0m 1s/step - accuracy: 0.5434 - loss: 0.4479
[32m-[0m[37m[0m [1m77s][0m 2s/step - accuracy: 0.5423 - loss: 0.4495 - val_accuracy: 0.5888 - val_loss: 0.4558
Epoch 2/15
[1m 1/17][0m [32m-[0m[37m-----[0m [1m5s][0m 331ms/step - accuracy: 0.4375 - loss: 0.6832
[32m-[0m[37m-----[0m [1m3s][0m 245ms/step - accuracy: 0.5078 - loss: 0.6098
[32m-[0m[37m-----[0m [1m3s][0m 244ms/step - accuracy: 0.5538 - loss: 0.5487
[32m-[0m[37m-----[0m [1m3s][0m 244ms/step - accuracy: 0.5677 - loss: 0.5234
[32m-[0m[37m-----[0m [1m2s][0m 244ms/step - accuracy: 0.5729 - loss: 0.5066
[32m-[0m[37m-----[0m [1m2s][0m 244ms/step - accuracy: 0.5738 - loss: 0.4967
[32m-[0m[37m-----[0m [1m2s][0m 244ms/step - accuracy: 0.5766 - loss: 0.4893
[32m-[0m[37m-----[0m [1m2s][0m 245ms/step - accuracy: 0.5788 - loss: 0.4839
[32m-[0m[37m-----[0m [1m1s][0m 245ms/step - accuracy: 0.5812 - loss: 0.4798
[32m-[0m[37m-----[0m [1m1s][0m 245ms/step - accuracy: 0.5809 - loss: 0.4779
[32m-[0m[37m-----[0m [1m1s][0m 245ms/step - accuracy: 0.5813 - loss: 0.4750
[32m-[0m[37m-----[0m [1m1s][0m 245ms/step - accuracy: 0.5830 - loss: 0.4720
[32m-[0m[37m-----[0m [1m0s][0m 245ms/step - accuracy: 0.5834 - loss: 0.4697
[32m-[0m[37m-----[0m [1m0s][0m 245ms/step - accuracy: 0.5843 - loss: 0.4674
[32m-[0m[37m-----[0m [1m0s][0m 245ms/step - accuracy: 0.5845 - loss: 0.4656
[32m-[0m[37m-----[0m [1m0s][0m 245ms/step - accuracy: 0.5855 - loss: 0.4634
[32m-[0m[37m[0m [1m0s][0m 241ms/step - accuracy: 0.5855 - loss: 0.4620
[32m-[0m[37m[0m [1m5s][0m 294ms/step - accuracy: 0.5855 - loss: 0.4607 - val_accuracy: 0.5588 - val_loss: 0.3720
Epoch 3/15

Training and Validation Loss for Fold 4



Training and Validation Accuracy for Fold 4



Best Hyperparameters for Fold 4:

```
{'conv_filters': 64, 'conv_activation': 'relu', 'unitsD1': 288, 'unitsD2': 160}
```

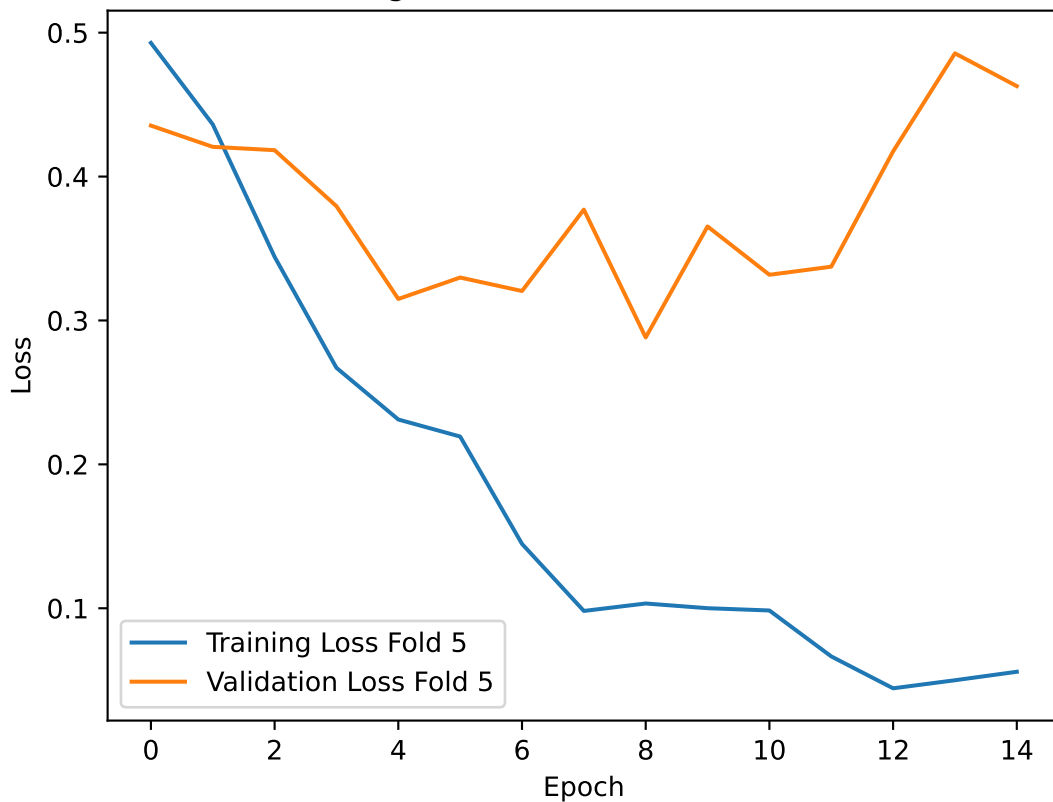
Best Trial Score for Fold 4: 0.8225000500679016

Search: Running Trial #1

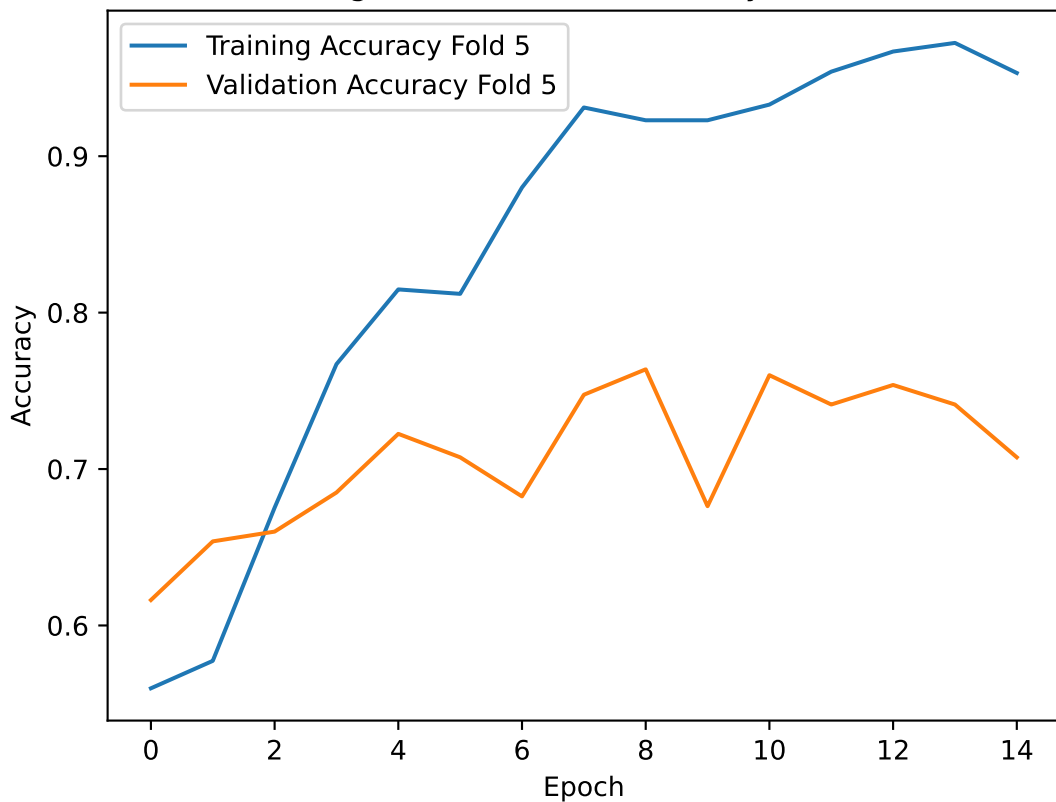
Value	Best Value So Far	Hyperparameter
64	64	conv_filters
relu	relu	conv_activation
288	288	unitsD1
160	160	unitsD2

Epoch 1/15
[1m 1/17][0m [32m-[0m[37m-----[0m [1m10:00][0m 38s/step - accuracy: 0.5625 - loss: 0.4752
[32m-[0m[37m-----[0m [1m3s][0m 252ms/step - accuracy: 0.5938 - loss: 0.4688
[32m-[0m[37m-----[0m [1m3s][0m 251ms/step - accuracy: 0.6389 - loss: 0.4288
[32m-[0m[37m-----[0m [1m3s][0m 251ms/step - accuracy: 0.6335 - loss: 0.4402
[32m-[0m[37m-----[0m [1m3s][0m 251ms/step - accuracy: 0.6193 - loss: 0.4509
[32m-[0m[37m-----[0m [1m2s][0m 251ms/step - accuracy: 0.6072 - loss: 0.4579
[32m-[0m[37m-----[0m [1m2s][0m 250ms/step - accuracy: 0.5995 - loss: 0.4627
[32m-[0m[37m-----[0m [1m2s][0m 250ms/step - accuracy: 0.5930 - loss: 0.4660
[32m-[0m[37m-----[0m [1m2s][0m 250ms/step - accuracy: 0.5888 - loss: 0.4679
[32m-[0m[37m-----[0m [1m1s][0m 250ms/step - accuracy: 0.5855 - loss: 0.4690
[32m-[0m[37m-----[0m [1m1s][0m 250ms/step - accuracy: 0.5811 - loss: 0.4718
[32m-[0m[37m-----[0m [1m1s][0m 250ms/step - accuracy: 0.5783 - loss: 0.4737
[32m-[0m[37m-----[0m [1m1s][0m 250ms/step - accuracy: 0.5754 - loss: 0.4752
[32m-[0m[37m-----[0m [1m0s][0m 250ms/step - accuracy: 0.5726 - loss: 0.4764
[32m-[0m[37m-----[0m [1m0s][0m 250ms/step - accuracy: 0.5702 - loss: 0.4773
[32m-[0m[37m-----[0m [1m0s][0m 250ms/step - accuracy: 0.5682 - loss: 0.4780
[32m-[0m[37m[0m [1m0s][0m 766ms/step - accuracy: 0.5665 - loss: 0.4787
[32m-[0m[37m[0m [1m69s][0m 2s/step - accuracy: 0.5651 - loss: 0.4793 - val_accuracy: 0.6450 - val_loss: 0.4160
Epoch 2/15
[1m 1/17][0m [32m-[0m[37m-----[0m [1m4s][0m 309ms/step - accuracy: 0.5938 - loss: 0.4402
[32m-[0m[37m-----[0m [1m3s][0m 250ms/step - accuracy: 0.6172 - loss: 0.4277
[32m-[0m[37m-----[0m [1m3s][0m 249ms/step - accuracy: 0.6545 - loss: 0.3893
[32m-[0m[37m-----[0m [1m3s][0m 249ms/step - accuracy: 0.6549 - loss: 0.3834
[32m-[0m[37m-----[0m [1m2s][0m 249ms/step - accuracy: 0.6465 - loss: 0.3853
[32m-[0m[37m-----[0m [1m2s][0m 249ms/step - accuracy: 0.6377 - loss: 0.3869
[32m-[0m[37m-----[0m [1m2s][0m 249ms/step - accuracy: 0.6339 - loss: 0.3881
[32m-[0m[37m-----[0m [1m2s][0m 249ms/step - accuracy: 0.6319 - loss: 0.3901
[32m-[0m[37m-----[0m [1m1s][0m 249ms/step - accuracy: 0.6323 - loss: 0.3913
[32m-[0m[37m-----[0m [1m1s][0m 249ms/step - accuracy: 0.6318 - loss: 0.3925
[32m-[0m[37m-----[0m [1m1s][0m 249ms/step - accuracy: 0.6320 - loss: 0.3931
[32m-[0m[37m-----[0m [1m1s][0m 249ms/step - accuracy: 0.6327 - loss: 0.3933
[32m-[0m[37m-----[0m [1m0s][0m 249ms/step - accuracy: 0.6330 - loss: 0.3935
[32m-[0m[37m-----[0m [1m0s][0m 249ms/step - accuracy: 0.6329 - loss: 0.3936
[32m-[0m[37m-----[0m [1m0s][0m 249ms/step - accuracy: 0.6330 - loss: 0.3938
[32m-[0m[37m-----[0m [1m0s][0m 249ms/step - accuracy: 0.6327 - loss: 0.3943
[32m-[0m[37m[0m [1m0s][0m 245ms/step - accuracy: 0.6319 - loss: 0.3951
[32m-[0m[37m[0m [1m8s][0m 472ms/step - accuracy: 0.6312 - loss: 0.3958 - val_accuracy: 0.6538 - val_loss: 0.3809
Epoch 3/15

Training and Validation Loss for Fold 5



Training and Validation Accuracy for Fold 5



Best Hyperparameters for Fold 5:

```
{'conv_filters': 256, 'conv_activation': 'relu', 'unitsD1': 288, 'unitsD2': 32}
```

Best Trial Score for Fold 5: 0.768750011920929

Search: Running Trial #1

Value	Best Value So Far	Hyperparameter
256	256	conv_filters
relu	relu	conv_activation
32	32	unitsD1
160	160	unitsD2

Epoch 1/15
[1m 1/17][0m [32m][0m][37m-----][0m [1m10:09][0m 38s/step - accuracy: 0.3438 - loss: 0.5149
[32m][0m][37m-----][0m [1m3s][0m 242ms/step - accuracy: 0.4375 - loss: 0.4913
[32m][0m][37m-----][0m [1m3s][0m 240ms/step - accuracy: 0.4826 - loss: 0.4757
[32m][0m][37m-----][0m [1m3s][0m 240ms/step - accuracy: 0.4948 - loss: 0.4757
[32m][0m][37m-----][0m [1m2s][0m 240ms/step - accuracy: 0.5033 - loss: 0.4758
[32m][0m][37m-----][0m [1m2s][0m 241ms/step - accuracy: 0.5054 - loss: 0.4781
[32m][0m][37m-----][0m [1m2s][0m 241ms/step - accuracy: 0.5104 - loss: 0.4793
[32m][0m][37m-----][0m [1m2s][0m 241ms/step - accuracy: 0.5139 - loss: 0.4800
[32m][0m][37m-----][0m [1m1s][0m 241ms/step - accuracy: 0.5170 - loss: 0.4808
[32m][0m][37m-----][0m [1m1s][0m 241ms/step - accuracy: 0.5184 - loss: 0.4817
[32m][0m][37m-----][0m [1m1s][0m 241ms/step - accuracy: 0.5217 - loss: 0.4811
[32m][0m][37m-----][0m [1m1s][0m 241ms/step - accuracy: 0.5242 - loss: 0.4815
[32m][0m][37m-----][0m [1m0s][0m 241ms/step - accuracy: 0.5270 - loss: 0.4818
[32m][0m][37m-----][0m [1m0s][0m 241ms/step - accuracy: 0.5290 - loss: 0.4823
[32m][0m][37m-----][0m [1m0s][0m 241ms/step - accuracy: 0.5299 - loss: 0.4832
[32m][0m][37m-----][0m [1m0s][0m 241ms/step - accuracy: 0.5307 - loss: 0.4841
[32m][0m][37m][0m [1m0s][0m 777ms/step - accuracy: 0.5305 - loss: 0.4851
[32m][0m][37m][0m [1m70s][0m 2s/step - accuracy: 0.5303 - loss: 0.4861 - val_accuracy: 0.5675 - val_loss: 0.4544
Epoch 2/15
[1m 1/17][0m [32m][0m][37m-----][0m [1m5s][0m 358ms/step - accuracy: 0.4062 - loss: 0.5148
[32m][0m][37m-----][0m [1m3s][0m 242ms/step - accuracy: 0.5469 - loss: 0.4213
[32m][0m][37m-----][0m [1m3s][0m 242ms/step - accuracy: 0.5868 - loss: 0.3894
[32m][0m][37m-----][0m [1m3s][0m 242ms/step - accuracy: 0.5944 - loss: 0.3914
[32m][0m][37m-----][0m [1m2s][0m 243ms/step - accuracy: 0.5980 - loss: 0.3922
[32m][0m][37m-----][0m [1m2s][0m 243ms/step - accuracy: 0.5982 - loss: 0.3983
[32m][0m][37m-----][0m [1m2s][0m 243ms/step - accuracy: 0.5988 - loss: 0.4035
[32m][0m][37m-----][0m [1m2s][0m 243ms/step - accuracy: 0.6001 - loss: 0.4075
[32m][0m][37m-----][0m [1m1s][0m 243ms/step - accuracy: 0.6025 - loss: 0.4099
[32m][0m][37m-----][0m [1m1s][0m 243ms/step - accuracy: 0.6029 - loss: 0.4125
[32m][0m][37m-----][0m [1m1s][0m 243ms/step - accuracy: 0.6033 - loss: 0.4147
[32m][0m][37m-----][0m [1m1s][0m 243ms/step - accuracy: 0.6043 - loss: 0.4160
[32m][0m][37m-----][0m [1m0s][0m 243ms/step - accuracy: 0.6051 - loss: 0.4169
[32m][0m][37m-----][0m [1m0s][0m 243ms/step - accuracy: 0.6062 - loss: 0.4173
[32m][0m][37m-----][0m [1m0s][0m 243ms/step - accuracy: 0.6071 - loss: 0.4176
[32m][0m][37m-----][0m [1m0s][0m 243ms/step - accuracy: 0.6081 - loss: 0.4176
[32m][0m][37m][0m [1m0s][0m 238ms/step - accuracy: 0.6077 - loss: 0.4180
[32m][0m][37m][0m [1m6s][0m 333ms/step - accuracy: 0.6074 - loss: 0.4184 - val_accuracy: 0.6075 - val_loss: 0.4558
Epoch 3/15

Average Validation F1-Score across all folds:

0.6505494505494506

Average Validation MSE across all folds:

0.38998788237571713

Average Validation Accuracy across all folds:

0.6771999971071879

Best Hyperparameters Found:

Best Fold: 4

Best Trial Score: 0.8225000500679016

Best Hyperparameters: {'conv_filters': 64, 'conv_activation': 'relu', 'unitsD